Workshop No. 1 Systems Analysis

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1. Proyect

Local Volunteer Coordination Platform: Matching volunteers with local community organizations and social impact projects. Skills-based matching algorithms, availability scheduling, and impact tracking system. Partnership with community organizations and social development initiatives.

2. System overview and scope definition

2.1. System Description

The system is a digital platform designed to connect volunteers with tutoring and educational support projects aimed at children and adolescents in local communities in Bogotá, Colombia. The goal is to optimize the assignment of volunteers based on:

- Academic skills (math, reading, English, technology)
- Teaching experience
- Availability
- Geographic location
- The system seeks to improve academic performance and reduce school dropout rates through efficient coordination

2.2. System Limits

The system includes:

- Registration and management of volunteer profiles
- Registration of educational institutions and projects
- Competency-based matching algorithm
- Schedule programming
- Monitoring and measuring academic impact
- Report generation

2.3. Main Components

The system consists of the following fundamental components:

Academic skills (math, reading, English, technology): The specific knowledge that the volunteer possesses in certain areas of learning is taken into account.

Teaching experience: This refers to the volunteer's previous experience in educational activities.

It may include: (Formal teaching experience, previous tutoring, academic mentoring, work with children or adolescents).

Availability: This indicates the days and times when the volunteer is available to provide service.

Volunteer location: Refers to the place where the volunteer lives or where they are willing to provide their services.

Volunteer Registration and Profiling Module: Responsible for collecting information on academic background, teaching experience, availability, and location.

Educational Institutions and Projects Registration Module: Where the academic needs, schedules, and characteristics of the students are defined.

Matching Algorithm: logical engine that calculates compatibility between volunteers and projects according to defined criteria

Scheduling System: coordinates schedules and assignments, avoiding availability conflicts.

Impact Monitoring and Measurement System: records sessions completed and generates academic improvement indicators.

Central Database: stores and manages all the information of the system. These components interact with each other to ensure efficient allocation.

2.4. Purpose of the System

The Community Education Volunteer Coordination Platform was created in response to the need to improve access to academic support in vulnerable communities. In many local contexts, educational institutions have limited human resources and personalized support, which affects school performance and increases the risk of dropouts. Likewise, there is a population of university students and professionals willing to volunteer, but without a structured mechanism to facilitate their effective involvement in specific educational projects. The system is therefore located at the intersection between the supply of volunteer talent and the demand for academic support, functioning as a technological mediator that optimizes this relationship.

The main purpose of the system is to optimize the connection between volunteers and tutoring projects, maximizing educational impact through an automated and structured process.

Specifically, the system seeks to:

- Reduce assignment times
- Improve compatibility between skills and needs
- Ensure effective schedule coordination

In terms of systems engineering, the purpose is to transform a manual, disorganized, and poorly measurable process into an efficient, traceable, and optimizable digital system.

2.5. Applicability Environment

The system operates in a local socio-educational environment characterized by:

- Educational institutions with limited resources
- Vulnerable communities with academic reinforcement needs
- Volunteers with university or technical training
- Basic technological infrastructure (internet access and digital devices)

External factors that influence the operating environment include:

- Availability of connectivity
- Academic calendars
- Socioeconomic conditions of students and institutions
- Institutional policies of participating organizations

The system interacts with this environment by adapting to variations in the supply of volunteers, educational demand, and logistical conditions, maintaining its function as a digital coordination platform.

2.6. Data collection plan and execution

This section describes the strategy designed to obtain first-hand information about the needs, expectations, and actual conditions of the system under analysis. Given the project's focus on coordinating volunteers for tutoring in communities in Bogotá, D.C., data collection is mainly oriented toward potential volunteers, mostly university students, as the central population of interest.

2.7. Methodology

A digital survey will be used as the primary data collection method. This digital survey will be designed in Google Forms and distributed via WhatsApp, targeting mainly university students at the Francisco José de Caldas District University. It will be distributed through academic groups and student volunteer networks. The digital survey will be structured as follows:

Thematic Block	Questions
Volunteer Profile	1. What is your field of study or profession? 2. In which subjects or academic areas do you consider yourself competent to teach?
Motivations and expectations	3. What would motivate you to participate in an educational volunteering program?
Previous volunteering experience	4. Have you previously carried out tutoring, mentoring or teaching activities?
Availability and logistics	5. How many hours per week would you be willing to dedicate to tutoring? 6. What days or time slots are you available? 7. Would you prefer to provide support in person, virtually, or both?

Why do we use them?
Because we need quantifiable data on:

- Skills
- Availability
- Interests

Organizational needs
The matching algorithm depends directly on this information.
What did we do?
We designed a form with questions such as:

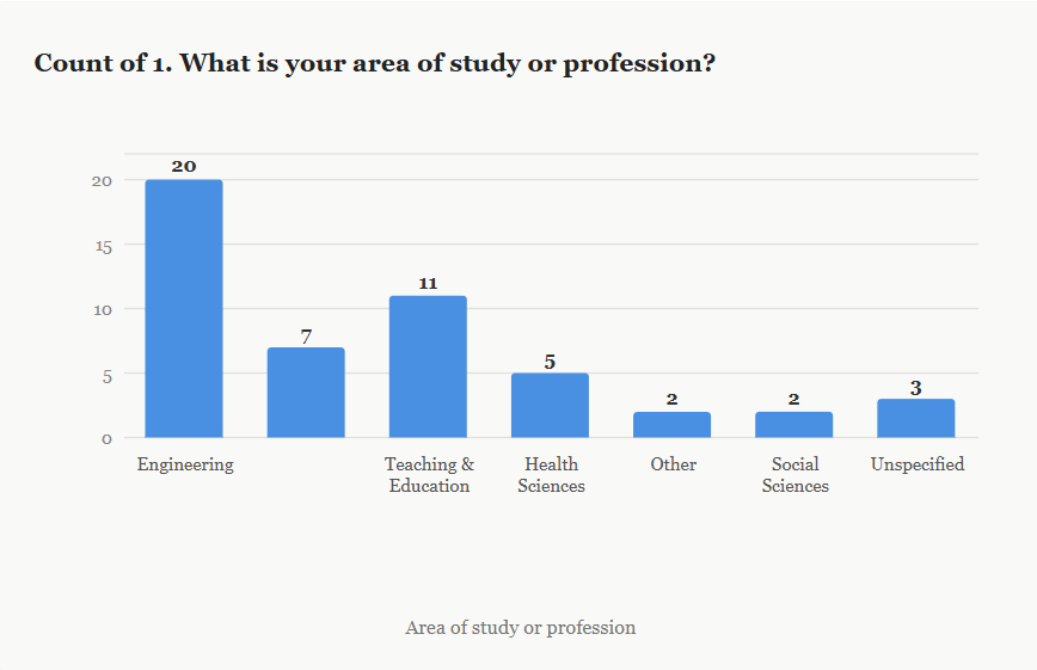
- What are your main skills?
- How many hours per week can you dedicate?
- What time slots are you available?
- What type of projects interest you?

We use Google Forms to conduct a data collection survey, the results of which are automatically stored in an Excel file. Participants' responses are completely anonymous, thus ensuring the confidentiality of their information.

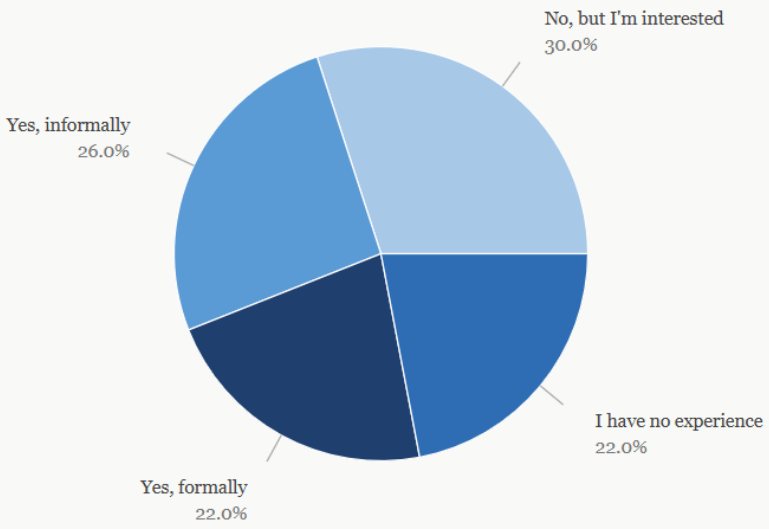
This is our survey:
Volunteer Profile Survey Community School Support Platform

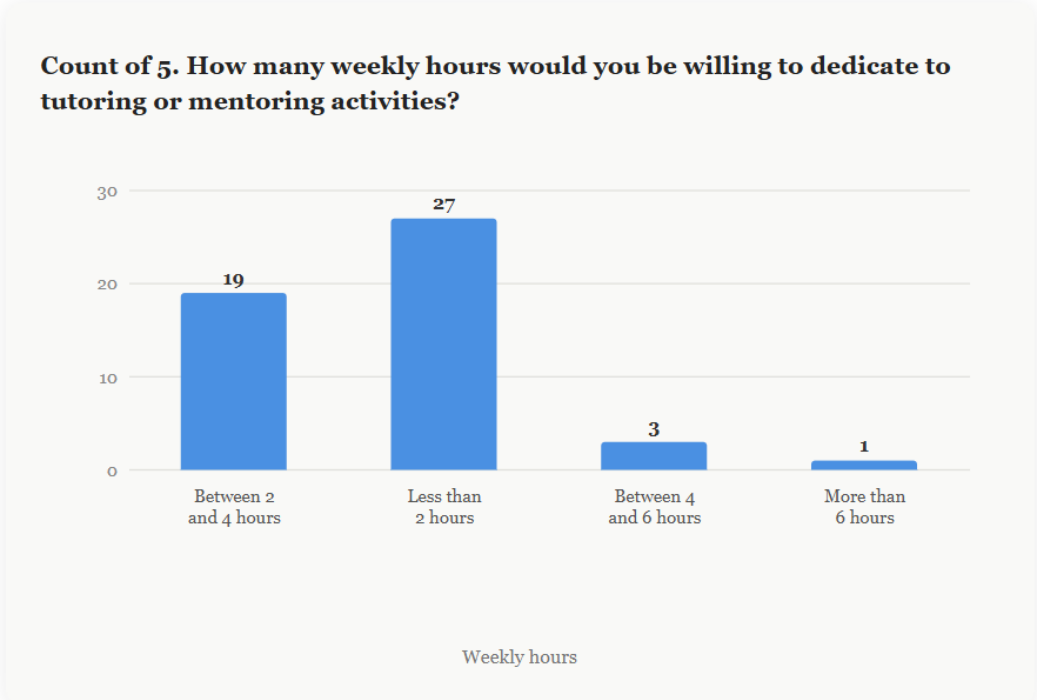
results:
Excel Results

2.8. Diagrams

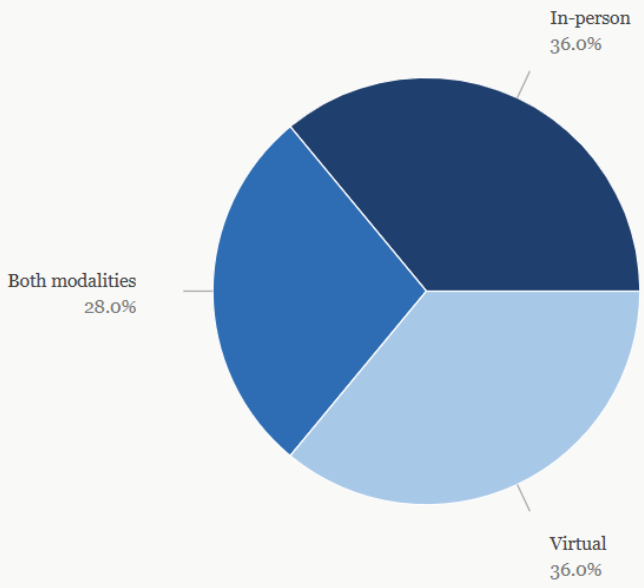

Count of 1. What is your area of study or profession?
20
15
10
5
0
20
7
11
5
2
2
3
Engineering
Teaching & Education
Health Sciences
Other
Social Sciences
Unspecified
Area of study or profession

Count of 4. Have you carried out tutoring activities, mentoring, or peer support?




Count of 5. How many weekly hours would you be willing to dedicate to tutoring or mentoring activities?
30
20
10
0
19
27
3
1
Between 2 and 4 hours
Less than 2 hours
Between 4 and 6 hours
More than 6 hours
Weekly hours

Count of 7. Would you prefer to provide the support in-person, virtually, or both?



3. Systems analysis report

The system connects volunteers with educational institutions that require academic support for students from local communities. For this purpose, different modules are used: volunteer registration, educational project registration, the matching algorithm, and the database that stores the information. Analyzing the system allows us to understand how these elements work together to facilitate volunteer assignment, improve tutoring organization, and generate a follow-up on the impact produced in the communities.

3.1. Elements and Relationships

The system is made up of several elements that interact with each other to enable the coordination of educational volunteering. Among the main elements are volunteers, educational institutions, school tutoring projects, the registration system, and the database.

Volunteers register information such as their academic competencies, teaching experience, schedule availability, and location. On the other hand, educational institutions register the academic support needs of their students.

The relationship between these elements is established through the matching algorithm, which analyzes the registered information to assign volunteers to the projects that best match their skills and availability.

3.2. Sensitivity and Variability

The system can be affected by different changes. An example is the number of available volunteers, the schedules in which they can participate, or the academic needs of the institutions.

If the number of volunteers increases, the system will have more options to make adequate assignments. Otherwise, it may be more difficult to cover all educational support needs.

For this reason, the system must be flexible enough to adapt to these changes and allow for proper coordination.

3.3. Complexity and Emergent Behavior

Within the system, emergent behaviors can appear thanks to the interaction between its different elements. Emergent behavior refers to situations or outcomes that arise when the system's components work together and that cannot always be predicted by analyzing each element separately.

In this case, when the number of registered volunteers increases or when educational institutions publish more tutoring projects, the system can generate new dynamics in tutor assignment. For example, better matches between volunteers and students may be formed, or academic support coverage may increase across different areas.

Emergent behavior allows us to observe how the system evolves over time and how the interaction between its components can generate results that improve the organization of educational volunteering.

3.4. Pattern Recognition

As the system collects information, it is possible to identify certain patterns in the data. For example, it is possible to detect which subjects have the highest demand for tutoring or the schedules during which volunteer availability is greatest.

These patterns help improve the organization of the system and support decision-making. They also help identify areas where greater volunteer participation is needed.

In this way, data analysis contributes to optimizing the performance of the platform.

3.5. Final Evaluation and Conclusions

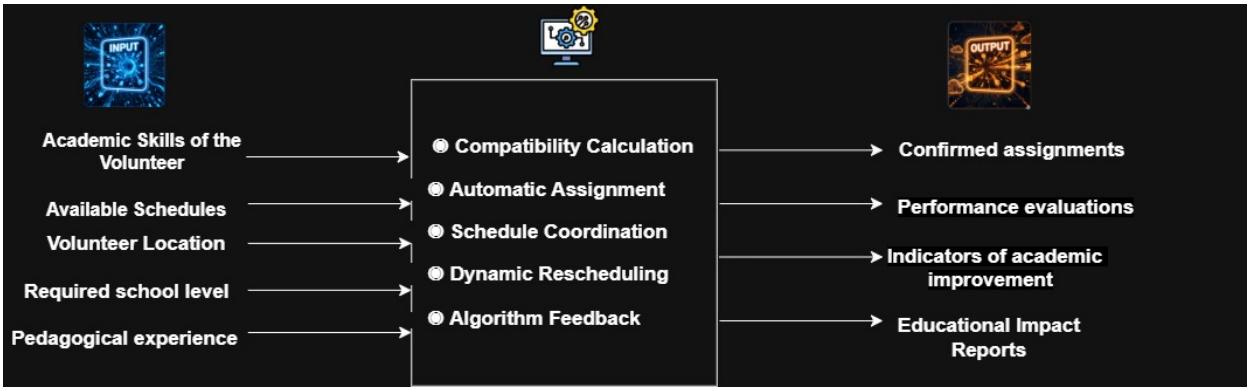
The system functions as a tool that facilitates coordination between volunteers and educational institutions requiring academic support. The interaction between the different system components allows for a more efficient organization of the tutor assignment process.

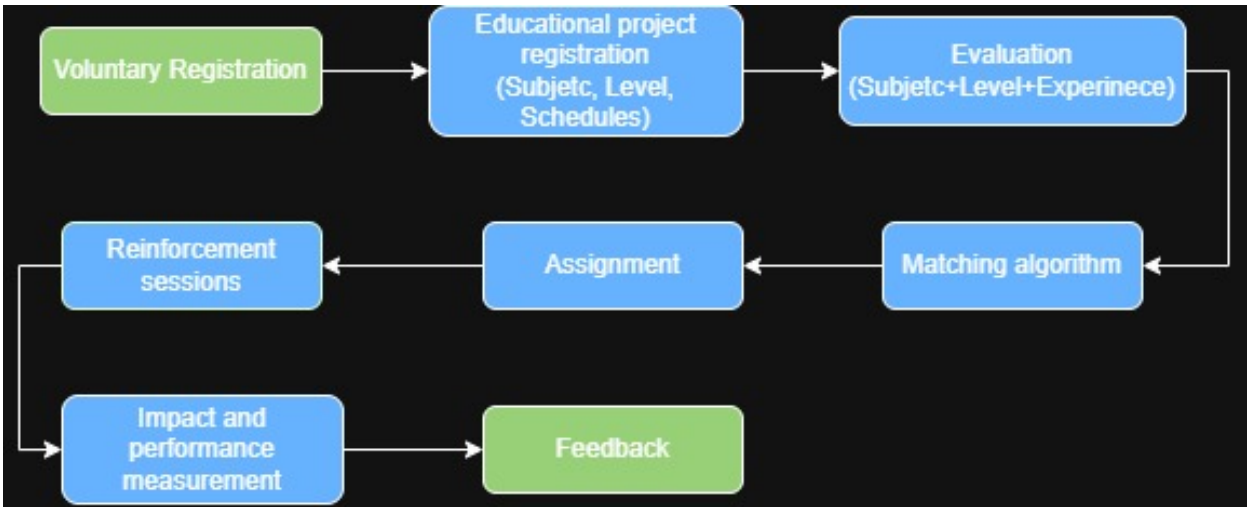
It is also identified that factors such as volunteer availability, institutional needs, and available schedules directly influence the functioning of the system. It is therefore important that the registered information is clear and up to date in order to achieve better assignment outcomes.

In conclusion, the proposed platform can contribute to improving the organization of volunteering by more efficiently connecting the supply of volunteers with the demand for academic support. Thus, the system can generate a positive impact on the learning process of students in these communities.

4. Visual presentation of the system

4.1. Diagrams





5. GitHub

Link GitHub